

ALI AHAD

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ABOUT

I am a Software Engineer at Google focusing on Security for Emergent AI Platforms. I hold a PhD in Electrical and Computer Engineering with a focus on systems and software security, specializing in program analysis, reverse engineering, and forensic techniques. Core Focus: *Security for Emergent AI Platforms* • *Program Analysis & Reverse Engineering* • *Systems & Software Security* • *Decompilation & Binary Hardening*

EDUCATION

- University of Maryland** – College Park, MD Defended April 2026
- PhD in Electrical and Computer Engineering
- University of Virginia** – Charlottesville, VA August 2023
- MS in Computer Science GPA: 4.0/4.0
- Lahore University of Management Sciences** – Lahore, Pakistan June 2020
- BS in Computer Science GPA: 3.9/4.0

EXPERIENCE

- Google** – Sunnyvale, CA August 2026 – Present
- Software Engineer*
- Security for Emergent AI Platforms.
- University of Maryland** – College Park, MD January 2024 – June 2026
- Research Assistant*
- Implement and evaluate novel malware analysis and information flow tracking methods; advance reverse engineering technologies to detect hidden malicious code and information leaks in real-world systems.
 - Developed FreePart, a partitioning technique for secure data processing (ASPLOS '24), and DeARM, an LLM-driven framework for improving ARM64 decompilation that outperforms SOTA decompilers by 10.2%.
- NVIDIA** – Santa Clara, CA May 2024 – August 2024
- System Software Intern, Data Center System Security*
- Enhanced TPM security for data center systems by implementing and validating DMTF/TCG TPM-over-SPDM protocols, prototyped on Virtual TPM (vTPM) for secure measurements and communications.
- NVIDIA** – Santa Clara, CA September 2023 – January 2024
- Software Intern, Security*
- Engineered a TPM-based attestation solution for server platform security, implementing EK/AK provisioning, RIM generation, and evidence reporting using the RATS architecture.
- University of Virginia** – Charlottesville, VA August 2020 – August 2023
- Research Assistant*
- Published four papers in top-tier venues in Security/Systems (CCS '21, FSE '21, S&P '22, S&P '23).
 - Led a cross-institutional project resulting in a first-author S&P '23 publication; mentored two undergraduate interns (Amazon Summer '22 & Appian Summer '23), guiding research and project development.

PUBLICATIONS

- ["FreePart: Hardening Data Processing Software via Framework-based Partitioning and Isolation"](#): Ali Ahad, Gang Wang, Chung Hwan Kim, Suman Jana, Zhiqiang Lin, and Yonghwi Kwon, *In Proc. of the 29th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS '24)*.
- ["PyFET: Forensically Equivalent Transformation for Python Binary Decompilation"](#): Ali Ahad, Chijung Jung, Ammar Askar, Doowon Kim, Taesoo Kim, and Yonghwi Kwon, *In Proc. of the 44th IEEE Symposium on Security and Privacy (S&P '23)*.
- ["SwarmFlawFinder: Discovering and Exploiting Logic Flaws of Swarm Algorithms"](#): Chijung Jung, Ali Ahad, Yuseok Jeon, and Yonghwi Kwon, *In Proc. of the 43rd IEEE Symposium on Security and Privacy (S&P '22)*.
- ["Forensic Analysis of Configuration-based Attacks"](#): Muhammad Adil Inam, Wajih Ul Hassan, Ali Ahad, Adam Bates, Rashid Tahir, Tianyin Xu, and Fareed Zaffar, *In Proc. of the 29th Network and Distributed System Security Symposium (NDSS '22)*.
- ["Swarmbug: Debugging Configuration Bugs in Swarm Robotics"](#): Chijung Jung, Ali Ahad, Jinho Jung, Sebastian Elbaum, and Yonghwi Kwon, *In Proc. of 29th ACM SIGSOFT International Symposium on the Foundations of Software Engineering (FSE '21)*.
- ["Spinner: Automated Dynamic Command Subsystem Perturbation"](#): Meng Wang, Chijung Jung, Ali Ahad, and Yonghwi Kwon, *In Proc. of 28th ACM Conference on Computer and Communications Security (CCS '21)*.

RELEVANT COURSES

- **Program Analysis**: Software Analysis, Program Analysis, Compilers
- **Security**: Mobile & IoT Security, Network Security & Privacy, Cyber Forensics
- **Systems**: Computer Architecture, Operating Systems, Digital CMOS VLSI Design
- **Machine Learning**: Intro. to Artificial Intelligence, Machine Learning, Information Theory
- **Networks**: Internet Infrastructure, Network-Centric Computing

TECHNOLOGY

- **Languages**: Python, C, C++, Rust, BASH, Golang, Javascript, Dart
- **Frameworks**: PyTorch, TensorFlow, HuggingFace Transformers, PEFT/LoRA, LLVM, Pandas, React, Flask
- **Reverse Engineering**: Ghidra, IDA, Angr, Uncompyle6, Decompyle3
- **Software Testing**: American Fuzzy Lop (AFL), KLEE, LibFuzzer
- **Infrastructure**: Git, Linux, Docker, SLURM, Wireshark

SKILLS

- **Security & Systems**: Systems & Software Security, Reverse Engineering, Program & Information Flow Analysis, Cyber Forensics, Secure System Design, Threat Modeling, Malware Analysis, Vulnerability Assessment
- **AI/ML & Program Analysis**: LLM Fine-Tuning & Specialization, Binary Analysis, Static & Dynamic Analysis, Algorithm Design & Optimization, Control Flow Analysis, Model Evaluation & Benchmarking
- **Systems Engineering**: Large-Scale Tool Development, Automated Testing & Fuzzing, Software Debugging
- **Research & Communication**: Cross-Functional Collaboration, Technical Roadmapping, Scientific Writing